

USAI GLOBAL AI HACKATHON

Winning Strategy Guide

A comprehensive, criterion-by-criterion playbook for dominating the High School track. From qualifier preparation to final submission, every strategy is mapped directly to the INFORMS-aligned judging rubric for maximum scoring impact.

High School Track
AI for Everyday Good
Community Challenge
June 2026

PREPARED BY Z.AI

Table of Contents

1. Hackathon Overview and Winning Analysis	2
1.1 What Makes This Hackathon Different	2
1.2 Judging Criteria Deep Dive	3
2. Qualifier Preparation (June 7-10)	4
2.1 Likely Qualifier Prompt Categories	4
2.2 Qualifier Pro Tips	5
3. Challenge Direction Selection	5
3.1 Community vs. Environment Comparison	5
3.2 Recommendation: Choose Community	6
4. Winning Project Concept: ClearPath AI	7
4.1 Problem Statement	7
4.2 Target Users	7
4.3 Solution Overview	8
5. AI Architecture Design	8
5.1 System Architecture	8
5.2 Data Flow	9
5.3 Technology Stack (All Free Tools)	10
6. Responsible AI Design	10
6.1 Human-in-the-Loop Design	10
6.2 Risk Identification and Mitigation	11

6.3 Submission-Ready Responsible AI Answers	11
7. Build Phase Strategy (June 14-21)	12
7.1 Build Priorities (What to Build First)	13
8. Submission Strategy	13
8.1 Pitch Video Structure (3-5 Minutes)	13
8.2 Project Description Template	14
8.3 AI Architecture Explanation (600 chars)	14
8.4 Tools and Data Disclosure	15
9. Scoring Maximization Playbook	15
9.1 Problem Understanding and Context	15
9.2 AI/Analytics Reasoning	16
9.3 Solution Design and Architecture	16
9.4 Impact and Decision Value	16
9.5 Responsibility, Ethics, and Limits	17
10. Pre-Submission Checklist	17

1. Hackathon Overview and Winning Analysis

The USAII Global AI Hackathon 2026 is not just another coding competition. It is a structured, two-phase innovation program that evaluates teams on the quality of their thinking, not just their code. Understanding this distinction is the single most important factor that separates winners from the rest of the field. While many teams will rush to build impressive technical demos, the judges are trained to look for something far more nuanced: the depth of your problem understanding, the intentionality of your AI approach, and the rigor of your responsible design choices. This guide provides a complete, step-by-step strategy to maximize your score across every single judging criterion.

1.1 What Makes This Hackathon Different

Three features set this hackathon apart from typical student competitions, and understanding each one is critical to your success. First, the two-phase quality design means that you must pass an AI Readiness Qualifier before you can even start building. This qualifier is not a trivia quiz; it is an evaluation of how well your team can think critically about AI solutions, demonstrating problem understanding, AI reasoning, and responsible AI awareness. Second, the INFORMS-aligned evaluation framework means that projects are judged on analytical reasoning and decision value, not just technical polish. A well-reasoned project with a simple but thoughtful AI approach will outscore a flashy demo with shallow reasoning every time. Third, the emphasis on responsible AI is not a checkbox at the end; it is woven into every part of the evaluation, from the qualifier through the final judging. Teams that treat ethics as an afterthought will be penalized, while teams that integrate responsible design into their core architecture will be rewarded.

1.2 Judging Criteria Deep Dive

Every decision you make during the hackathon should be mapped to one of the five judging criteria. Think of these criteria as the scoring rubric that the judges hold in their hands. Here is what each criterion really means and how to maximize your score:

Criterion	What Judges Look For	How to Maximize Score
Problem Understanding and Context	Do you clearly understand who is affected, why the problem matters, and the decision context where your solution will be used?	Include specific user personas, cite real statistics, describe the exact moment of frustration your user faces, and explain why existing solutions fall short.
AI/Analytics Reasoning	Is AI an appropriate and intentional approach, not just bolted on for trendiness?	Explain why non-AI approaches (simple search, static database) would fail. Show that AI is necessary for the problem, not decorative. Describe your AI capability specifically.
Solution Design and Architecture	Is the solution logically structured with clear inputs, AI components, and outputs working together?	Draw a clear architecture diagram. Show data flow from input to output. Explain each component and why it exists. Be specific about what AI model or technique you use and why.
Impact and Decision Value	Does the solution help someone make a better decision or take a meaningful action?	Define a specific decision your user makes. Show how your AI improves that decision. Measure impact with concrete metrics (even estimated). Tell a before/after story.
Responsibility, Ethics, and Limits	Did you consider risks, ethical implications, and limitations of your solution?	Identify at least one realistic risk with a concrete mitigation. Define one decision the AI does NOT make. Be transparent about limitations. Show crisis safeguards.

Table 1. Judging criteria breakdown with scoring maximization strategies

2. Qualifier Preparation (June 7-10)

The AI Readiness Qualifier is your gateway to the hackathon. It consists of approximately 8 prompts about a hypothetical scenario, evaluating your problem understanding, AI thinking, responsible AI awareness, and clarity of communication. The qualifier takes about 30 minutes and does not require building anything. It is automatically scored using an AI-assisted rubric. Based on the scoring rubric and the judging criteria, here are the likely prompt categories and how to answer each one for maximum points.

2.1 Likely Qualifier Prompt Categories

Prompt Category	What It Tests	Winning Answer Strategy
1. Problem Identification	Can you clearly articulate a real problem with specific affected users?	Name a specific, concrete problem. Identify exact users (not "everyone"). Cite a real statistic or example. Explain why the problem persists despite existing solutions.
2. User Analysis	Do you understand who is affected and their decision context?	Create a specific persona (name, age, situation). Describe the exact moment they struggle. Explain their emotional state and barriers. Show empathy without being vague.
3. AI Approach Justification	Is AI the right tool, not just a trendy add-on?	Explain why simpler approaches (keyword search, static FAQ) would fail. Identify the specific AI capability needed (NLP understanding, pattern recognition, classification). Be precise about what the AI does.
4. Input/Output Design	Can you define what goes into and comes out of your AI system?	Specify exact inputs (text description, location, language preference). Define exact outputs (ranked resource list, simplified eligibility, action steps). Show the transformation between input and output.
5. Data Strategy	Do you know what data your AI needs and where it comes from?	Identify real, publicly available data sources. Explain how you would handle missing data. Consider data quality and bias. Mention synthetic data if real data is unavailable.
6. Responsible AI Awareness	Can you identify risks and design mitigations?	Identify one realistic, specific risk (not generic "AI bias"). Propose one concrete, implementable mitigation (not "we will be careful"). Show you understand the harm that could come from your system failing.
7. Human-in-the-Loop	Do you know where AI should NOT make decisions?	Identify one specific decision that must remain with a human. Explain why (what could go wrong if AI decided). Show the boundary between AI assistance and AI autonomy.

8. Communication Clarity	Can you explain complex ideas simply and persuasively?	Use short, clear sentences. Avoid jargon. Use analogies. Structure your answer with clear logic (problem, then approach, then reasoning). Be specific, not abstract.
-----------------------------	--	--

Table 2. Likely qualifier prompt categories with winning answer strategies

2.2 Qualifier Pro Tips

- Be specific, not generic.** "People in rural areas without reliable internet" is infinitely better than "people who need help." The AI scorer is trained to identify vague, boilerplate answers and penalize them. Every claim you make should be grounded in a concrete example or data point.
- Show AI reasoning, not just AI enthusiasm.** Explain WHY AI is the right tool for your specific problem. If a simple database search could solve it, then AI is not justified and you will lose points. The qualifier rewards intentional, well-reasoned AI choices over flashy but unnecessary ones.
- Demonstrate responsible AI thinking early.** Do not wait for the ethics prompt to show responsible AI awareness. Weave ethical considerations into your problem analysis and approach design. For example, when describing your users, acknowledge that some may be in crisis situations and explain how your design accounts for that.
- Write clearly and concisely.** The qualifier is timed, so practice writing structured answers before June 7. Use the format: one clear claim, then supporting evidence, then a brief conclusion. Avoid long rambling paragraphs that bury your key insight.
- Prepare examples in advance.** Before the qualifier opens, research 2-3 real community organizations (United Way, 211.org, local food banks) and understand their challenges. Having concrete examples ready will make your answers stand out from teams who only speak in abstractions.

3. Challenge Direction Selection

The high school track offers two challenge directions: Community (Help is Hard to Find) and Environment (Make Climate Action Local and Real). Your choice of direction will fundamentally shape your project, so this decision deserves careful analysis. While both directions offer strong opportunities, our recommendation is the Community direction for the reasons detailed below.

3.1 Community vs. Environment Comparison

Factor	Community: Help is Hard to Find	Environment: Climate Action Local and Real
--------	---------------------------------	--

Problem Relatability	Universal. Everyone has experienced confusing systems or not knowing where to turn for help. Judges will immediately empathize.	Strong but abstract. Climate concern is widespread, but personal experience with climate tools is less universal among teens.
AI Justification	Very strong. NLP is genuinely needed to understand free-text descriptions of need, match them to resources, and simplify jargon. Simple search cannot do this well.	Moderate. Some AI applications (image recognition for waste sorting, prediction models) are valid, but simpler rule-based tools often suffice for tracking/education.
Responsible AI Story	Extremely strong. Crisis detection, always-surface-hotlines, and human-in-the-loop for eligibility decisions create a powerful and authentic responsible AI narrative.	Moderate. Environmental tools have less immediate safety risk, making the responsible AI story less dramatic and harder to make compelling.
Impact Measurability	Clear. You can measure: resources found, time saved, crisis interventions triggered, user satisfaction.	Harder. Environmental impact is long-term and indirect. Short-term metrics (actions logged, pledges made) feel less substantial.
Data Availability	Excellent. 211.org, United Way, government databases, Crisis Text Line data. Rich, structured, publicly available.	Good. EPA data, weather APIs, waste databases exist but are often less structured and harder to integrate for a prototype.
Special Award Fit	Dual fit. Strong candidate for BOTH Best Responsible AI Design AND Best Social Impact. Crisis safeguards + helping vulnerable populations = compelling narratives for both awards.	Single fit. Strong for Best Social Impact but weaker for Best Responsible AI Design since ethical risks are lower and less dramatic.

Table 3. Side-by-side comparison of the two high school challenge directions

3.2 Recommendation: Choose Community

Based on the analysis above, the Community direction offers a significant competitive advantage for three interconnected reasons. First, the AI justification is stronger and more natural. When a person types "I lost my job and I do not know how to pay rent," understanding that sentence and matching it to rental assistance programs, food banks, and job training resources genuinely requires natural language processing. A simple keyword search on "job" would miss the rental assistance connection entirely. This makes your AI reasoning easy to defend and compelling to explain, which directly boosts your score on the AI/Analytics Reasoning criterion.

Second, the responsible AI narrative is more powerful and authentic. When you are building a tool that people in crisis might use, the stakes are real and the need for safeguards is obvious. Your decision to always display crisis hotline information regardless of AI classification is not a theoretical design choice; it is a life-saving feature. This gives you a natural, compelling answer to the Responsible AI judging criterion and makes you a strong candidate for the Best Responsible AI Design special award.

Third, the social impact is more immediate and measurable. When someone uses your tool and finds a food bank they did not know about, that is a tangible, countable impact. Environmental impact, while critically

important, is inherently long-term and harder to demonstrate convincingly in a one-week hackathon. This directness strengthens your Impact and Decision Value score and positions you well for the Best Social Impact special award. By choosing Community, you can realistically compete for both special awards simultaneously, doubling your chances of bringing home prize money.

4. Winning Project Concept: ClearPath AI

Based on our analysis of the judging criteria, challenge requirements, and competitive landscape, we recommend building **ClearPath AI**, an intelligent community resource navigator that helps people find support through plain-language conversation. This concept is specifically designed to score maximum points across all five judging criteria and compete for both special awards.

4.1 Problem Statement

Across the United States and around the world, millions of people need help with basic needs like food, housing, healthcare, and legal assistance, but they cannot find it. The problem is not that resources do not exist; the problem is that the systems connecting people to resources are broken. Government websites are written in bureaucratic jargon. Eligibility requirements are confusing and often contradictory across programs. Phone lines have hold times measured in hours. Community organizations operate in silos, each with their own database, their own intake process, and their own terminology. For someone already stressed, overwhelmed, or in crisis, navigating this maze feels impossible.

Consider Maria, a 34-year-old single mother who just lost her job. She needs help with rent, food for her children, and healthcare coverage. Right now, she would need to visit at least four different websites, fill out four different applications, and decipher eligibility requirements written in legal language. She might spend days on this process, and she might still miss programs she qualifies for simply because she did not know the right keywords to search for. ClearPath AI changes this by letting Maria describe her situation in her own words and instantly receiving a clear, personalized guide to the help available to her.

4.2 Target Users

Persona	Situation	Core Need	Barrier Today
Maria (34, single mother)	Lost job, needs rent help, food, and healthcare for her kids	A single guide to all programs she qualifies for, in simple language	Four different websites, confusing eligibility jargon, does not know what to search for
James (17, high school student)	Experiencing housing instability, needs to find shelter and food assistance	Discreet, mobile-friendly way to find nearby help without embarrassment	Embarrassed to ask for help in person, does not know what programs exist for teens

Priya (68, retiree)	Needs Medicare help and transportation to medical appointments	Simple, step-by-step instructions for complex government programs	Overwhelmed by Medicare website complexity, limited tech literacy
Community workers	Social workers, librarians, teachers who help others find resources	Quick, reliable way to look up resources for the people they serve	Memorizing dozens of programs, outdated referral lists, no unified tool

Table 4. Target user personas for ClearPath AI

4.3 Solution Overview

ClearPath AI is a conversational community resource navigator that works in three simple steps. First, the user describes their situation in plain language, typing something like "I lost my job and I cannot afford rent this month" or "my family needs food help near me." Second, the AI analyzes their description using natural language processing to extract their specific needs (housing, food, healthcare, employment, legal, education, or crisis), detect urgency level, and identify relevant eligibility factors. Third, ClearPath AI returns a personalized results page showing matched resources ranked by relevance, simplified eligibility requirements translated into plain English, step-by-step action instructions for each resource, and crisis hotline information that is always visible regardless of the AI classification result.

The key innovation of ClearPath AI is not that it replaces human judgment, but that it removes the navigation barrier that prevents people from getting to the right human in the first place. By transforming a confusing, multi-step research process into a single conversational interaction, ClearPath AI dramatically reduces the time, knowledge, and emotional energy required to find help. The system is designed with a mobile-first interface, acknowledging that many people in need primarily access the internet through their phones.

5. AI Architecture Design

The architecture of ClearPath AI is designed to be clear, explainable, and appropriate for a high school team building with free tools during a one-week hackathon. Every component has a specific purpose that maps directly to a judging criterion, ensuring that when judges ask "why did you build it this way," you have a compelling, well-reasoned answer.

5.1 System Architecture

Component	Function	Technology (Free)	Judging Criterion Addressed
Input Interface	User describes needs in plain language via text chat or form	HTML/CSS/JavaScript with responsive mobile-first design	Problem Understanding (shows user empathy)

Need Extraction Engine	Classifies user description into need categories (housing, food, healthcare, etc.)	Hugging Face Transformers zero-shot classification (free, open-source)	AI Reasoning (shows intentional AI use)
Urgency Detector	Identifies crisis-level situations (suicidal ideation, domestic violence, acute need)	Keyword + sentiment analysis with rule-based escalation triggers	Responsibility (crisis safeguard)
Resource Matcher	Matches extracted needs to relevant community resources using semantic similarity	Sentence embeddings via Hugging Face (sentence-transformers, free)	Architecture (clear data flow)
Eligibility Simplifier	Translates complex eligibility requirements into simple language	Template-based simplification with pre-written plain English explanations	Impact (actionable output)
Action Plan Generator	Creates step-by-step instructions for contacting each resource	Structured template filling with resource-specific contact info	Decision Value (clear next steps)
Crisis Safety Layer	Always displays crisis hotlines regardless of AI classification	Hard-coded safety component that cannot be bypassed by AI output	Responsibility (non-negotiable safeguard)
Output Display	Shows ranked resources, simplified eligibility, action steps, and crisis info	React or vanilla JavaScript with clear visual hierarchy	All criteria (complete user experience)

Table 5. ClearPath AI system architecture with judging criterion alignment

5.2 Data Flow

The data flow through ClearPath AI follows a clear, linear pipeline that is easy to explain to judges and demonstrates intentional design at every stage. When a user submits their description, the text first passes through the Need Extraction Engine, which uses a pre-trained zero-shot classification model to identify which categories of need are present (housing, food, healthcare, legal, employment, education, crisis). The model outputs a confidence score for each category, and any category above a set threshold is flagged as relevant.

Simultaneously, the Urgency Detector analyzes the same input text for crisis indicators. This is a rule-based system that checks for specific high-risk keywords and patterns (such as mentions of self-harm, domestic violence, or acute safety threats) combined with sentiment analysis to detect extreme distress. If any crisis indicators are detected, the system immediately elevates the response priority and ensures crisis resources are prominently displayed at the top of the results. Importantly, even when no crisis indicators are detected, crisis hotlines are still displayed at the bottom of every response. This is a deliberate design choice: the cost of showing an unnecessary hotline number is zero, but the cost of missing someone in crisis could be catastrophic.

The extracted needs then flow to the Resource Matcher, which compares the need categories against a database of community resources using semantic similarity. Each resource in the database has a textual description, and the matcher uses sentence embeddings to find resources whose descriptions most closely align with the user expressed needs. This is where the AI genuinely adds value beyond a simple keyword search: a user who writes "I cannot afford my apartment" will be matched to rental assistance programs even if those programs are described in the database as "housing stabilization services," because the semantic meaning is captured, not just the literal words.

5.3 Technology Stack (All Free Tools)

Layer	Tool	Why This Choice
Frontend	HTML/CSS/JavaScript or Next.js	Free, widely known, mobile-responsive. Next.js adds routing and SSR if team has React experience.
AI/NLP	Hugging Face Transformers (zero-shot classification, sentence-transformers)	Free, open-source, no API key needed, runs locally. Industry-standard NLP toolkit.
Backend	Python Flask or Node.js Express	Simple, lightweight, free. Flask is ideal for AI/NLP integration since Hugging Face runs in Python.
Resource Database	211.org API + United Way directory + synthetic local data	211.org provides real community resource data via API. Synthetic data fills gaps for local areas without API coverage.
Hosting	Vercel (frontend) + Render or Railway (backend)	Free tiers available. Vercel is ideal for Next.js. Render offers free Python hosting.
Version Control	GitHub	Free, required for submission, enables collaboration.

Table 6. Technology stack using only free tools

6. Responsible AI Design

Responsible AI is not an add-on for ClearPath AI; it is a core architectural principle. The hackathon explicitly rewards teams that integrate ethical considerations into their design, and the Best Responsible AI Design special award (\$250 + scholarship) is directly achievable with the right approach. Here is how ClearPath AI addresses responsibility, ethics, and limits across every dimension the judges will evaluate.

6.1 Human-in-the-Loop Design

The single most important responsible AI design decision in ClearPath AI is this: **the AI never makes the final determination about whether someone qualifies for a resource.** This is the answer to the required Human-in-the-Loop submission question, and it is a powerful one. When a user searches for help, ClearPath

AI suggests resources and simplifies eligibility language, but it never tells a user "you qualify" or "you do not qualify." The actual eligibility determination is always made by the resource provider (the caseworker, the government agency, the nonprofit intake specialist). The AI role is to help the user get to the right provider faster, not to replace the provider judgment.

This design choice is justified by a clear risk analysis: if the AI incorrectly told someone they qualified for a program they do not actually qualify for, that person might make life decisions based on false expectations (such as not seeking alternative help, or spending time on an application that will be denied). Conversely, if the AI incorrectly told someone they do not qualify, they might give up on seeking help they are entitled to. Either type of error could cause real harm to vulnerable people. By keeping the eligibility determination in human hands and positioning the AI as a navigation assistant, ClearPath AI avoids both risks entirely.

6.2 Risk Identification and Mitigation

Risk	Severity	Mitigation Design	Why This Works
Misclassifying urgency and failing to surface crisis resources	Critical	Crisis hotlines are ALWAYS displayed regardless of AI classification. Separate "I need immediate help" button bypasses AI entirely.	Eliminates the possibility of someone in crisis not seeing help. Cost of extra info is zero; cost of missing crisis is catastrophic.
AI provides incorrect eligibility information	High	AI never makes final eligibility determinations. All eligibility info is labeled "Contact the provider to confirm." Simplified language is template-based, not AI-generated.	Prevents users from making decisions based on AI errors. Human verification is built into the process, not optional.
Bias in resource recommendations	Medium	Resource database includes all available resources, not just AI-ranked ones. Users can browse full catalog. Ranking is transparent and explainable.	AI suggestions are helpful, not restrictive. Users always have access to the complete picture, not just what the AI chose.
Privacy concerns with sensitive personal information	High	No user accounts required. No data stored after session ends. Input is processed in real-time and discarded. Clear privacy notice on first screen.	People seeking help should not have to create accounts or share personal data. Session-based design respects privacy by default.
Over-reliance on AI for critical decisions	Medium	Prominent disclaimer: "This tool helps you find resources. It does not replace professional advice." AI output is clearly labeled as suggestions, not directives.	Sets clear expectations about AI capabilities. Users understand they are getting navigation help, not professional guidance.

Table 7. Risk identification and mitigation design for ClearPath AI

6.3 Submission-Ready Responsible AI Answers

The hackathon submission requires two specific responsible AI answers with strict character limits. Here are your draft answers, carefully crafted to fit within the limits while maximizing impact:

Human-in-the-Loop Design (500 chars):

ClearPath AI never determines eligibility for resources. The AI suggests and simplifies, but final eligibility decisions remain with human providers (caseworkers, agencies). If AI incorrectly said "you qualify," users might skip alternatives; if it said "you don't qualify," users might give up seeking help they need. This boundary protects vulnerable people from AI errors with real consequences.

Responsible AI Guardrail (500 chars):

Risk: Misclassifying urgent situations, causing someone in crisis to not see emergency help. Mitigation: Crisis hotline info (988 Suicide Hotline, Crisis Text Line, domestic violence hotline) is ALWAYS displayed regardless of AI classification. A prominent "I need immediate help" button bypasses all AI processing and goes directly to crisis resources. This design accepts that AI can fail and ensures safety nets work even when it does.

7. Build Phase Strategy (June 14-21)

The build phase is one week long, and the most successful teams are those who prioritize ruthlessly. You do not need production-ready code, perfect UI, or enterprise infrastructure. You need a clear demonstration of how your AI works, evidence of AI capability, and a compelling explanation of your approach. Here is a day-by-day plan optimized for maximum judging impact.

Day	Date	Priority Tasks	Deliverable by End of Day
Day 1	June 14	Attend kickoff livestream. Finalize project scope. Set up GitHub repo. Build resource database (at least 30 resources with descriptions). Set up Flask backend skeleton.	Working repo with database, API skeleton, and clear README
Day 2	June 15	Implement Need Extraction Engine using Hugging Face zero-shot classification. Test with 10+ example inputs. Implement Urgency Detector with crisis keywords.	Working NLP pipeline that correctly classifies needs from text input
Day 3	June 16	Implement Resource Matcher using sentence-transformers for semantic similarity. Build Eligibility Simplifier with plain-English templates. Integrate all AI components.	End-to-end AI pipeline: input text to ranked resource list
Day 4	June 17	Build frontend interface (mobile-first chat or form UI). Connect frontend to backend API. Implement Crisis Safety Layer. Display results with clear visual hierarchy.	Working web app with full user experience

Day 5	June 18	Testing and refinement. Test with diverse scenarios (non-native English, crisis situations, vague descriptions). Fix bugs. Add disclaimers and privacy notice. Deploy to free hosting.	Deployed, tested app accessible via public URL
Day 6	June 19	Record pitch video (3-5 minutes). Write Devpost submission (description, design questions, tools/data disclosure). Create architecture diagram for demo.	Completed pitch video and written submission draft
Day 7	June 20	Final review of all submission components. Test demo link. Verify qualifier approval code. Get team feedback. Submit early (do not wait until 11:59 PM).	Submitted project with all required components

Table 8. Day-by-day build phase plan

7.1 Build Priorities (What to Build First)

If you run out of time, here is the priority order for what matters most to judges. Build these first and polish later:

Priority 1: Working AI pipeline. The single most important thing is showing that your AI actually works. A user types a description and gets relevant results. This is non-negotiable. Without a working AI component, you cannot demonstrate AI capability.

Priority 2: Crisis Safety Layer. The always-visible crisis hotlines and "I need immediate help" button. This is your strongest responsible AI story and directly contributes to the Best Responsible AI Design award.

Priority 3: Clear results display. Users must be able to understand the output. Simplified eligibility, action steps, and ranked resources. The interface does not need to be beautiful, but it must be clear.

Priority 4: Architecture documentation. A clear diagram showing how inputs flow through your AI components to produce outputs. Judges will reference this when evaluating your Solution Design.

Priority 5: Pitch video. A compelling 3-5 minute video that tells your story. This is often the first thing judges watch, and a strong video sets the tone for how they evaluate everything else.

8. Submission Strategy

Your submission is the only thing the judges see, so every component must be optimized for maximum impact. Here is how to craft each required component to score the highest possible points across all five judging criteria.

8.1 Pitch Video Structure (3-5 Minutes)

Section	Time	Content	Judging Criterion
Hook	0-30s	Start with a powerful statistic or story. "Every day, thousands of people who need help cannot find it, not because it does not exist, but because the system is broken." Show the problem viscerally.	Problem Understanding
Solution	30s-2 min	Introduce ClearPath AI. Explain why AI is needed (not just a database search). Walk through the architecture briefly. Highlight the responsible AI design choices.	AI Reasoning, Architecture, Responsibility
Demo	2min -3:30	Show the app working with a real example. Type a user description. Show the AI classifying needs, matching resources, and displaying results. Demonstrate the crisis safety feature.	Architecture, AI Reasoning
Impact	3:30-5min	Explain who benefits and how. Share before/after comparison. Discuss limitations honestly. End with a vision for what comes next.	Impact, Decision Value, Responsibility

Table 9. Pitch video structure with judging criterion alignment

8.2 Project Description Template

The Devpost project description is your written pitch. Structure it with these exact sections to ensure you address every judging criterion clearly. Use bold headers so judges can find what they need quickly:

What it does and why it matters: One paragraph explaining the problem (with a specific user story) and one paragraph explaining your solution. Lead with impact, not technology.

How we built it: Describe your architecture, AI approach, and technology choices. Include a brief mention of your responsible AI design. Show the data flow from input to output.

Challenges we ran into: Be honest and specific. "The zero-shot classification model initially struggled with vague descriptions like I just need help, so we added a clarification prompt" is much more impressive than "it was hard to code."

Accomplishments we are proud of: Highlight your responsible AI design, your working AI pipeline, and any creative solutions to tough problems. This is your chance to shine.

What we learned: Show intellectual growth. "We learned that AI classification confidence scores alone are not reliable enough for crisis detection, which is why we built a rule-based safety layer on top" demonstrates deep understanding.

What is next: Be realistic but ambitious. Mention partnerships with local nonprofits, expansion to more communities, or additional languages. This shows your project has legs beyond the hackathon.

8.3 AI Architecture Explanation (600 chars)

Here is your submission-ready AI Architecture Explanation, designed to fit within the 600-character limit while clearly addressing all four required elements (inputs, AI capability, processing, outputs):

Inputs: User describes their situation in plain text (e.g., "lost my job, need help with rent and food"). AI Capability: Zero-shot NLP classification extracts need categories (housing, food, healthcare, etc.); sentence-transformers compute semantic similarity for resource matching; rule-based urgency detection flags crisis situations. Processing: Input text is classified into need categories, matched against a database of community resources using semantic similarity, and assessed for urgency level. Outputs: Ranked resource list with simplified eligibility, step-by-step action plans, and always-visible crisis hotlines.

8.4 Tools and Data Disclosure

AI Tools Used (800 chars):

Hugging Face Transformers (free, open-source) - zero-shot classification and sentence-transformers for NLP; Python Flask (free, open-source) - backend API server; HTML/CSS/JavaScript (free) - frontend interface; GitHub Copilot (free for students) - code assistance and debugging; VS Code (free) - development environment; Google Colab (free) - model testing and experimentation; ChatGPT free tier - brainstorming and documentation assistance

Data Sources (800 chars):

211.org community resource directory API (public API, real service data); United Way resource database (publicly available listings); FindHelp.org open resource data (public directory of social services); Synthetic test scenarios generated using ChatGPT free tier (50 diverse user need descriptions for testing); Crisis hotline data from SAMHSA national hotline database (publicly available government data); Local resource data compiled from public city and county websites

9. Scoring Maximization Playbook

This section provides specific, actionable tactics to maximize your score on each of the five judging criteria. These are not generic tips; they are specific strategies tailored to the ClearPath AI concept and the hackathon evaluation framework.

9.1 Problem Understanding and Context

The judges want to see that you do not just understand the problem in the abstract, but that you understand the specific people affected, the specific moments of frustration they experience, and the specific reasons existing solutions fail them. To maximize your score on this criterion, you should lead every explanation with a specific user story, not a general statement. Instead of saying "people struggle to find resources," say "when Maria types lost my job into a government website search bar, she gets results about unemployment insurance forms, but not about the food bank three blocks from her house or the rental assistance program she qualifies for." The specificity makes the problem real and demonstrates deep understanding.

You should also cite real statistics to ground your problem statement. For example: "According to the Urban Institute, 1 in 4 Americans who qualify for safety net programs do not receive them, often because they do not know the programs exist or cannot navigate the application process." This shows you have done your research and are not just guessing that the problem exists. Additionally, explain the decision context: when someone needs help, they are making decisions under stress, with limited time, and often with shame or fear. Your solution must account for this emotional reality, not just the informational gap.

9.2 AI/Analytics Reasoning

This criterion asks whether AI is the right tool for your problem. The most convincing way to demonstrate this is to explain what would happen WITHOUT AI. If a simple keyword search or a static database could solve the problem equally well, then AI is not justified, and you will lose points. For ClearPath AI, the argument is straightforward: when someone writes "I cannot afford my apartment anymore," a keyword search for "apartment" would return apartment listings, not rental assistance programs. The semantic gap between how people describe their needs and how resources are officially categorized is exactly the kind of problem that NLP is designed to solve. You should make this argument explicitly in your submission and video.

Additionally, you should explain your specific AI capability with technical precision. Do not just say "we use AI to match people with resources." Say "we use zero-shot text classification to identify need categories from free-text descriptions, and sentence-transformers to compute semantic similarity between user needs and resource descriptions." This level of specificity shows the judges that you understand what your AI is actually doing and why you chose that particular approach over alternatives.

9.3 Solution Design and Architecture

The architecture criterion rewards clarity and logical structure. Your architecture diagram should be simple enough to explain in 30 seconds but detailed enough to show that every component has a purpose. Avoid the common mistake of creating an overly complex diagram with many arrows and boxes that looks impressive but is impossible to explain clearly. Instead, use a linear pipeline diagram: Input, then Need Extraction, then Urgency Detection (parallel), then Resource Matching, then Output. Each box should have a one-line description of what it does and what technology it uses. The crisis safety layer should be shown as a separate, always-active component that sits alongside the AI pipeline, not as part of it. This visual separation reinforces your responsible AI narrative.

9.4 Impact and Decision Value

To maximize your impact score, you need to clearly define the decision your user is making and show how your AI improves it. The decision is: "Which resource should I contact first, and what do I need to do to get help?" Without ClearPath AI, this decision is made blindly, through hours of searching across confusing websites. With ClearPath AI, the user has a ranked list of relevant resources, simplified eligibility information, and clear next steps. You should present this as a before/after comparison in your pitch video, showing the same user scenario with and without your tool. Even estimated metrics are powerful: "What

takes 2-3 hours of searching today could take 2-3 minutes with ClearPath AI." The key is to make the impact tangible and concrete, not abstract and theoretical.

9.5 Responsibility, Ethics, and Limits

This criterion is where ClearPath AI truly separates itself from the competition. Most teams will add a generic "we considered bias" statement to their submission. Your advantage is that responsible AI is baked into your architecture, not bolted on at the end. The three strongest points to emphasize are: first, your human-in-the-loop design where the AI never makes eligibility determinations (this is a concrete, implementable boundary, not a vague promise); second, your crisis safety layer that always displays hotline information regardless of AI output (this is a specific, testable safeguard, not a theoretical consideration); and third, your privacy-by-design approach where no user data is stored (this addresses a real concern for vulnerable populations who might fear government tracking or data breaches). Each of these points is specific, implementable, and demonstrates mature ethical thinking that will resonate with judges.

10. Pre-Submission Checklist

Use this checklist to ensure your submission is complete and optimized. Check every item before submitting, and submit at least 2 hours before the deadline to avoid last-minute technical issues.

Category	Item	Critical Notes
Required Components	Qualifier Approval Code (8 characters)	Without this, your submission is DISQUALIFIED. Double-check the code from your June 12 email.
Required Components	Project Title and Tagline (80 chars max)	Title should be descriptive and memorable. Tagline should be one clear sentence.
Required Components	Full Project Description	Must include all 6 sections: what it does, how built, challenges, accomplishments, learned, next.
Design Questions	AI Architecture Explanation (600 chars)	Cover all 4 elements: inputs, AI capability, processing, outputs. Be specific, not vague.
Design Questions	Human-in-the-Loop Design (500 chars)	Name ONE decision the AI does NOT make. Explain WHY a human must be in control.
Design Questions	Responsible AI Guardrail (500 chars)	Name ONE realistic risk. Show ONE concrete mitigation. Use the crisis hotline example.
Tools and Data	AI Tools Used (800 chars)	List ALL tools, specify free vs. paid. Disclose all AI assistance including coding help.
Tools and Data	Data Sources (800 chars)	List all datasets with links. Explain synthetic data generation. Be transparent.

Demo Materials	Pitch Video (3-5 minutes)	Must be 3-5 minutes. Follow Hook-Solution-Demo-Impact structure. Test the link works.
Demo Materials	Demo or Walkthrough Link	Live app, GitHub repo with README, or video walkthrough. PowerPoint-only is NOT accepted.
Administrative	Track and Challenge selected	High School track, Community: Help is Hard to Find direction.
Administrative	All team members listed	Every team member must be on the Devpost submission. 2-5 members required.

Table 10. Complete pre-submission checklist

This strategy guide is designed to give you every possible advantage in the USAII Global AI Hackathon 2026. The key insight throughout is that judges reward depth of thinking over technical flash. A team that deeply understands their problem, chooses AI intentionally, designs responsibly, and communicates clearly will outperform a team with fancier technology but shallower reasoning every time. Good luck, and build AI for good!